

Name: _____ Date: _____

Period: _____

Problems 1–6**1.** For each function, identify the amplitude and midline.

Function	Amplitude	Midline
$f(\theta) = 7 \sin(\theta) + 3$		
$g(\theta) = -5 \cos(\theta) - 4$		
$h(\theta) = \frac{3}{2} \sin(2\theta) + 1$		
$k(\theta) = -\cos(\pi\theta)$		

2. For each function, identify the period by first finding b .

(a) $f(\theta) = \sin(4\theta)$ $b = \underline{\hspace{2cm}}$ Period $= \frac{2\pi}{|b|} = \underline{\hspace{2cm}}$

(b) $g(\theta) = \cos\left(\frac{\pi}{3}\theta\right)$ $b = \underline{\hspace{2cm}}$ Period $= \underline{\hspace{2cm}}$

(c) $h(\theta) = 3 \sin\left(\frac{1}{4}\theta\right) - 2$ Period $= \underline{\hspace{2cm}}$

3. For each function, factor the argument and identify the phase shift. State the direction and amount.

(a) $f(\theta) = \sin(2\theta - \pi)$
Factor: $2\theta - \pi = 2(\theta - \underline{\hspace{2cm}})$ Phase shift: $\underline{\hspace{2cm}}$

(b) $g(\theta) = \cos\left(3\theta + \frac{\pi}{2}\right)$
Factor: $3\theta + \frac{\pi}{2} = 3\left(\theta + \underline{\hspace{2cm}}\right)$ Phase shift: $\underline{\hspace{2cm}}$

(c) $h(\theta) = \sin\left(\frac{1}{2}\theta - \frac{\pi}{4}\right)$
Factor: $\underline{\hspace{2cm}}$ Phase shift: $\underline{\hspace{2cm}}$

4. Identify all four features (amplitude, period, phase shift, midline) for each function.

(a) $f(\theta) = 4 \sin\left(3\left(\theta - \frac{\pi}{6}\right)\right) + 5$
Amplitude: $\underline{\hspace{2cm}}$ Period: $\underline{\hspace{2cm}}$ Phase shift: $\underline{\hspace{2cm}}$ Midline: $\underline{\hspace{2cm}}$

(b) $g(\theta) = -2 \cos(4\theta + \pi) - 3$
First factor: $4\theta + \pi = 4(\theta + \underline{\hspace{2cm}})$
Amplitude: $\underline{\hspace{2cm}}$ Period: $\underline{\hspace{2cm}}$ Phase shift: $\underline{\hspace{2cm}}$ Midline: $\underline{\hspace{2cm}}$

5. (AP-Style MC) Which of the following is the period of $h(\theta) = 5 \sin(3\theta - \pi) + 2$?

- (A) $\pi/3$
 (B) $2\pi/3$
 (C) 3π
 (D) π
 (E) 2π

Answer: $\underline{\hspace{2cm}}$

6. (AP-Style MC) The function $f(\theta) = a \sin(b\theta) + d$ has amplitude 6 and midline $y = -1$. Which of the following could be f ?

- (A) $f(\theta) = 6 \sin(\theta) + 1$
- (B) $f(\theta) = -6 \sin(\theta) - 1$
- (C) $f(\theta) = 6 \sin(\theta) - 7$
- (D) $f(\theta) = 3 \sin(2\theta) - 1$
- (E) $f(\theta) = 6 \sin(\theta) + 7$

Answer: _____

Extension optional

Extension (Optional): The function $p(\theta) = A \cos(B\theta) + D$ has maximum value 7 and minimum value -1 .

- (a) Find the amplitude A and midline D of p .

Amplitude $= \frac{\max - \min}{2} = \underline{\hspace{2cm}}$ Midline $D = \frac{\max + \min}{2} = \underline{\hspace{2cm}}$

- (b) If the period of p is 4π , find B .

- (c) Rewrite $p(\theta)$ as a sine function with an appropriate phase shift. Show all work. _____

Preview — Lesson 3.7: Sinusoidal Function Graphs

You can now read amplitude, period, phase shift, and midline directly from a sinusoidal equation. In Lesson 3.7 we'll go in both directions: from equation to graph (using key points), and from a graph back to the equation. Come ready to use today's parameters to plot and interpret complete sinusoidal graphs.